OPENING--AND-CLOSING

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Aim

To implement Opening and Closing using Python and OpenCV.

Software Required

1. Anaconda - Python 3.7
2. OpenCV

Algorithm:

Step1:

Import the necessary packages

Step2:

Create the Text using cv2.putText

Step3:

Create the structuring element

Step4:

Use Opening operation

Step5:

Use Closing Operation

Program:

```
# NAME :
# REF NO :

# Import libraries
import cv2
```

```python
import numpy as np
import matplotlib.pyplot as plt

# Read image in grayscale
image = cv2.imread("rose.png", 0)

# Create structuring element (kernel)
kernel = np.ones((5,5), np.uint8)

# ---------------- OPENING ----------------
opening = cv2.morphologyEx(image, cv2.MORPH_OPEN, kernel)

# ---------------- CLOSING ----------------
closing = cv2.morphologyEx(image, cv2.MORPH_CLOSE, kernel)

# ---------------- DISPLAY OUTPUT ----------------
titles = ["Original Image", "Opening", "Closing"]
images = [image, opening, closing]

plt.figure(figsize=(10,5))

for i in range(3):
    plt.subplot(1,3,i+1)
    plt.imshow(images[i], cmap='gray')
    plt.title(titles[i])
    plt.axis("off")

plt.tight_layout()
plt.show()
```

Output:

Display the input Image

Original Image

Display the result of Opening

Opening

Display the result of Closing

Closing

Result

Thus the Opening and Closing operation is used in the image using python and OpenCV.